S.T. YAU HIGH SCHOOL SCIENCE AWARD (ASIA) 2026 · PHYSICS

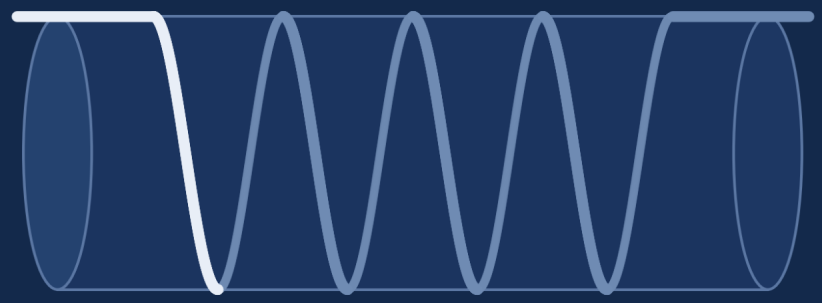
A Predictive Amplitude Law for the Vibration-Driven Slip Front in Capstan Friction

Presented by Jevesh Jain

Team Phy-236 · Jevesh Jain and Mrityunjay Gupta

Supervisor: Mr Gajendra Sethi · Jayshree Periwal International School, India

Field: tribology and contact mechanics



A four-wrap capstan contact: creeping region (bright) and gripping region

Background: a wrapped rope can restrain a large load

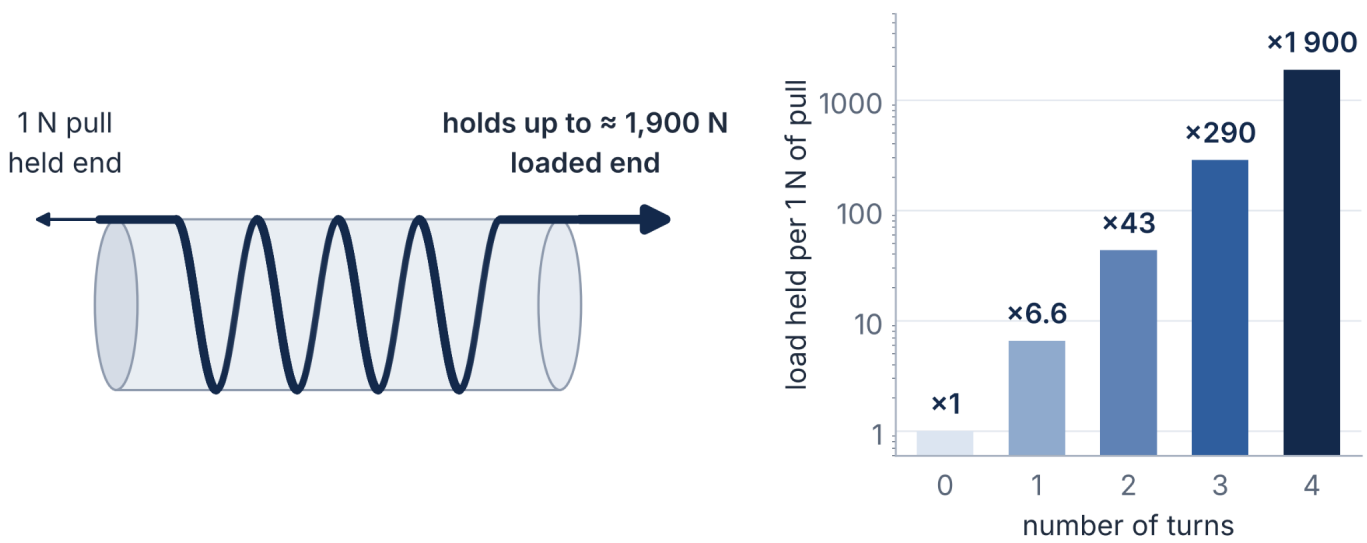


Figure 1 · A small holding force restrains a large load; the ratio multiplies with each turn
Source: schematic prepared for this presentation (μ = 0.30, report Table 2)

EULER'S CAPSTAN LAW (1762)

$$T_2 = T_1 e^{\mu\Theta}$$

T_1: holding tension · T_2: largest load that can be restrained · μ: friction coefficient · Θ: total wrap angle

Observation

A small force at one end of a wrapped rope restrains a much larger load at the other.

Mechanism

Tension presses the rope onto the drum, so friction rises with tension; each turn builds on the last.

Magnitude

Four turns at μ = 0.30: a 1 N holding force restrains up to about 1,900 N.

KEY POINT Each additional turn multiplies the load a rope can restrain: the capstan effect.

Motivation and relevance: capstans in service vibrate

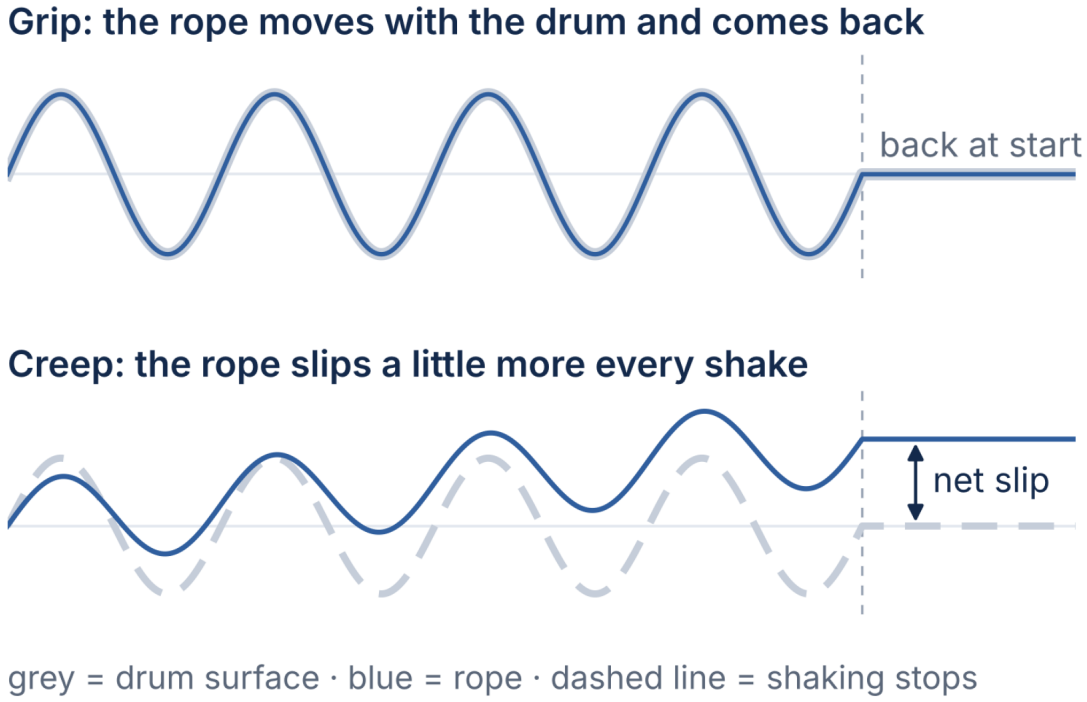


Figure 2 · Displacement of one rope element over four oscillation cycles: stick (top), creep (bottom)
Source: schematic for this talk; slip magnified about 100× for clarity

Applications

Ship winches, mooring fairleads, cranes and cable-driven robots: wrapped ropes on drums subject to vibration in service.

Observation

The rope does not slip as a whole. One region, however, creeps: it accumulates a small slip with every oscillation cycle.

Significance

Repeated micro-slip at 30–60 Hz causes fretting wear, concentrated in the creeping region of the rope.

KEY POINT The capstan law predicts how much load a rope can sustain, but not which region of it creeps.

Research question and hypothesis

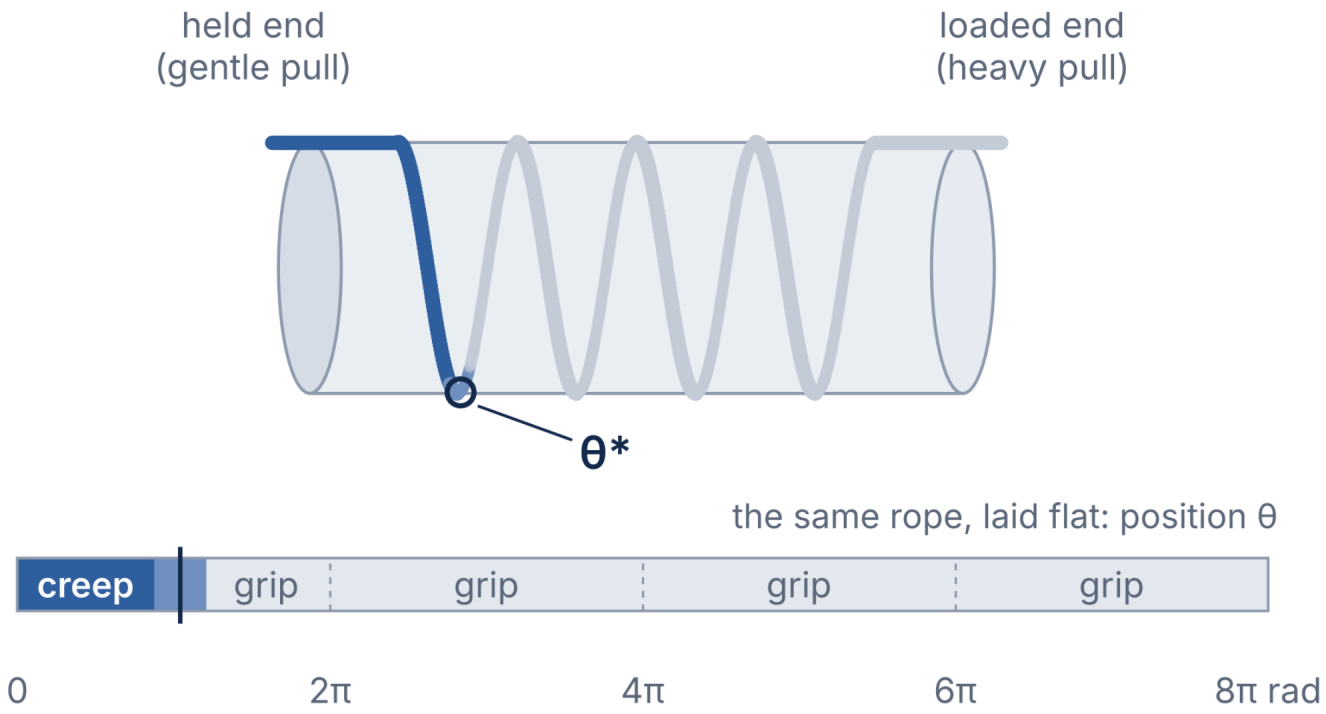


Figure 3 · Hypothesis: creep near the held end, stick beyond a transition position θ*
Source: schematic after report Fig. 1; the position of θ* is illustrative

Research question

Under capstan vibration, does creep extend over the whole contact or end at a definite position? If it ends, can that position be predicted before measurement?

Hypothesis

1. Creep begins at the held end, where tension, and hence frictional resistance, is lowest.
2. Creep ends at a definite transition position, θ*.
3. θ* increases with vibration amplitude, by an amount predictable in advance.

KEY POINT θ* is the position at which creep ends, measured from the held end (1 rad ≈ 12.5 mm of rope).

Key definitions used in this talk

θ

Position along the contact

Cumulative contact angle measured from the held end. 1 rad ≈ 12.5 mm of rope; four turns = 8π rad.

μ

Friction coefficients

Rope on aluminium: static μs = 0.30; reduced (slow-sliding) μred = 0.18.

T_1

Held-end tension

The lower of the two end tensions, about 1 N. The loaded end carries $T_2 = 5\ T_1$ throughout.

vc

Creep rate

Relative slip of a rope element per oscillation cycle (μm cycle^{-1}). Below 0.1 μm cycle^{-1} is classed as stick.

u_0

Vibration amplitude

Peak tangential displacement of the drum surface; reference value 0.26 mm.

θ*

Transition position

Where creep ends and stick begins: the quantity this work predicts and measures.

KEY POINT The central quantity of this work is θ*, the position at which creep ends.

Literature review: three foundations of this work

EULER (1762) [4]; EXTENSIONS [6–8, 10]

Capstan friction

$$T_2 = T_1 e^{\mu\Theta}$$

Key result: the tension a wrapped rope can sustain grows exponentially with wrap angle.

Role in this work: sets the tension, and hence the frictional resistance, at every point of the contact.

Limitation: predicts a single gross-slip limit for the whole contact; it does not locate partial slip.

CATTANEO (1938) [13]; MINDLIN (1949) [14]

Preslip contact stiffness

$$q_{\text{vib}} = \kappa_t u_0$$

Key result: below the friction limit, a contact deforms elastically with a finite tangential stiffness κt.

Role in this work: converts the imposed vibration displacement u_0 into a tangential force on the rope.

Limitation: derived for compact contacts, not for a rope wrapped over four turns.

LUBARDA (2015) [12]

Spatial slip on a static capstan

$$\theta_s = \frac{1}{\mu} \ln \frac{T_2}{T_1}$$

Key result: on a stationary drum under slowly increasing load, friction is mobilised first at the loaded end.

Role in this work: the static benchmark against which the vibration-driven case is compared.

Limitation: no vibration is considered, and contact stiffness plays no part.

KEY POINT The gap: no existing theory predicts where along a wrapped rope vibration-driven creep ends.

Closest prior work: static versus vibration-driven slip

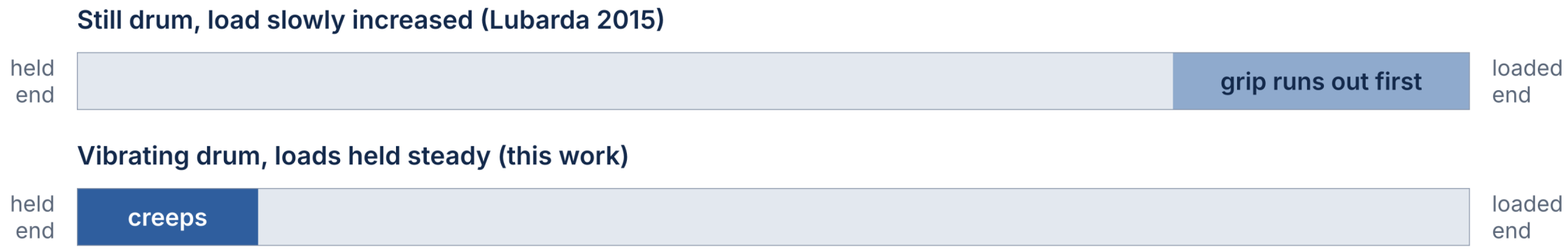


Figure 4 · The contact unwrapped, held end on the left: static drum versus vibrating drum
Source: schematic based on Lubarda (2015) [12] and this work

Origin of slip

Static drum: friction is mobilised first at the heavily loaded end. Vibrating drum: creep begins at the held end, where resistance is lowest.

Governing parameters

Static drum: load ratio and friction coefficient only. Vibrating drum: additionally, vibration amplitude and contact stiffness.

Driving mechanism

Belt-creep theories require a belt travelling round a pulley. Here the drum only oscillates, so vibration alone drives the creep.

KEY POINT Under vibration, slip originates at the opposite end, and its extent depends on contact stiffness.

Our innovation: novelty of this work

SYNTHESIS

Capstan friction
exponential tension profile along the wrap [4]

+

Preslip contact stiffness
elastic response of a contact before sliding [13, 14]

→

A predictive law for θ*
position at which vibration-driven creep ends

1

Predictive law

A closed-form expression for θ*.

2

Independent calibration

Every parameter measured separately; none fitted to θ*.

3

Spatial measurement

High-speed optical tracking of creep along all four wraps.

4

Model discrimination

Five competing spatial models compared by BIC.

5

Confirmatory tests

Five pre-registered predictions tested across 84 runs.

KEY POINT **This work extends capstan theory from a load limit to a predictive law for position, and tests it quantitatively.**

Theory (1 of 3): two competing forces on every piece of rope

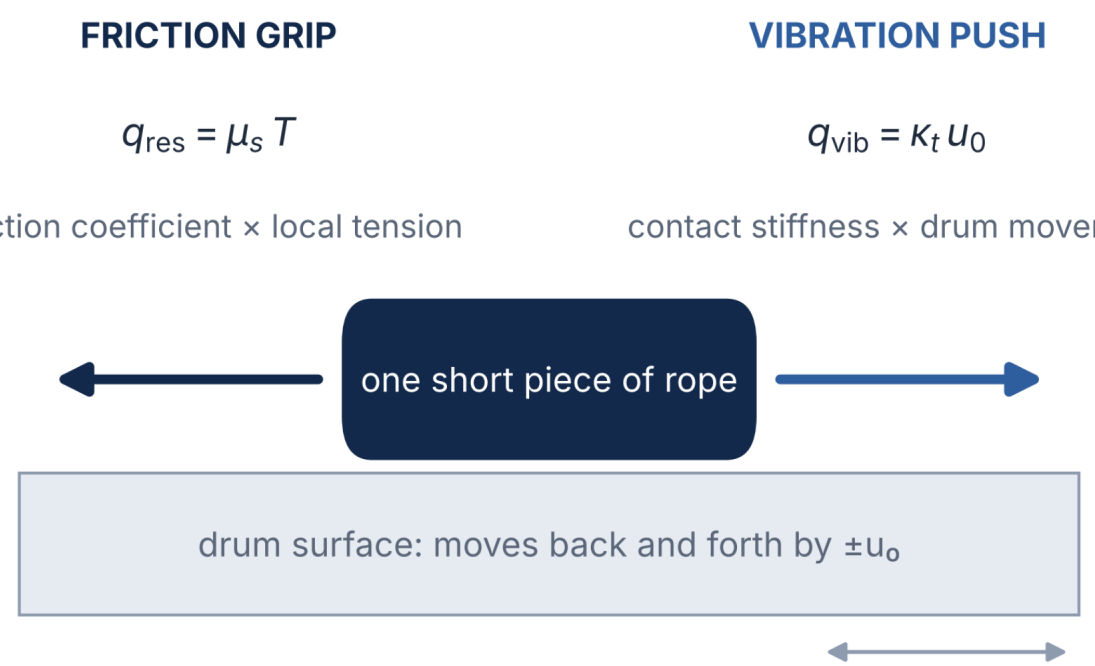


Figure 5 · The two forces acting on one short piece of rope in contact with the drum
Source: schematic prepared for this presentation

Vibration push

$q_{vib} = \kappa_t u_0$

The drum surface moves back and forth by the same distance u_0 (≈ 0.26 mm) all along the contact. Before slipping, the contact flexes like a stiff spring, so the drum drags the rope with force $\kappa t \times u_0$.

Friction grip

$q_{res} = \mu_s T_{eff}$

Friction can resist at most μs times the local rope tension, Teff, since tension presses the rope onto the drum.

The rule for each piece of rope

Push > grip: it slips slightly every cycle (creep). Grip > push: it holds (stick).

KEY POINT Each piece of rope creeps if the vibration push exceeds its friction grip, and holds otherwise.

Theory (2 of 3): why creep stops, and how we derived the law

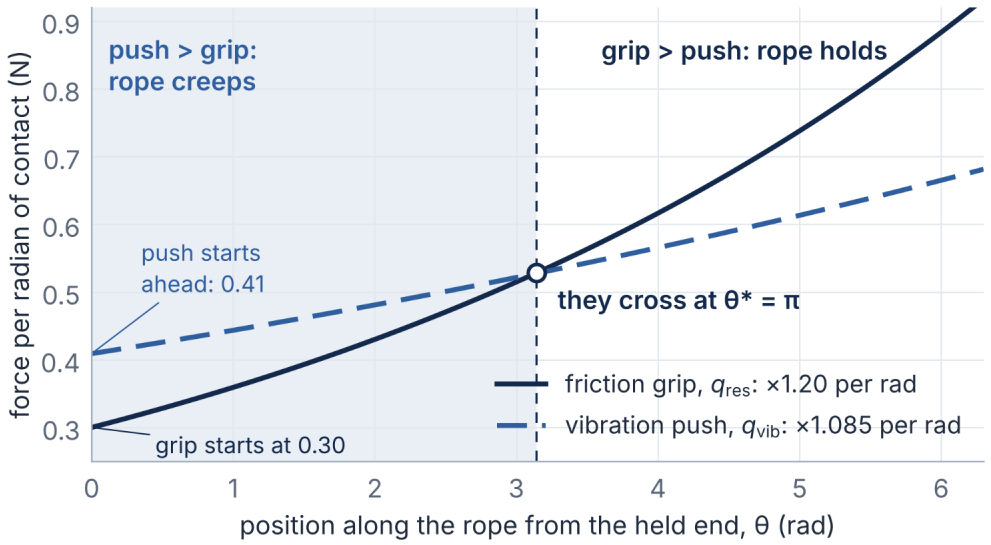


Figure 6 · Push and grip along the rope at the reference setting
Source: computed from the bench-calibrated constants (slide 13)

1 **Grip grows along the rope, as tension multiplies (capstan effect): ×1.20 per radian.**

$$q_{res}(\theta) = \mu_s T_1 e^{\mu_{red}\theta}$$

2 **Push grows more slowly: stiffness rises only as tension to the power α = 0.45, so ×1.085 per radian.**

$$q_{vib}(\theta) = \kappa_{ref} u_0 \left(\frac{T_1}{T_{ref}}\right)^{\alpha} e^{\alpha\mu_{red}\theta}$$

3 **Creep stops where push equals grip. Call that position θ*.**

$$\kappa_{ref} u_0 \left(\frac{T_1}{T_{ref}}\right)^{\alpha} e^{\alpha\mu_{red}\theta^*} = \mu_s T_1 e^{\mu_{red}\theta^*}$$

4 **Divide, take the natural logarithm and solve: Equation 1.**

$$\theta^* = \frac{1}{(1-\alpha)\,\mu_{red}} \ln\left(\frac{\kappa_{ref} u_0}{\mu_s T_{ref}^{\alpha} T_1^{1-\alpha}}\right)$$

KEY POINT Push starts ahead but grip grows faster, so they cross once: that crossing is θ*, given by Equation 1.

Theory (3 of 3): the law and its five predictions

how far push starts ahead of grip at the held end (push ÷ grip)

$$\theta^{*} = \frac{1}{(1-\alpha)\,\mu_{\mathrm{red}}} \ln\left(\frac{\kappa_{\mathrm{ref}}\,u_0}{\mu_s\,T_{\mathrm{ref}}^{\alpha}\,T_1^{1-\alpha}}\right)$$

1 ÷ how fast grip catches up with push, per radian

Equation 1 · The transition-position law; every constant measured on the bench beforehand
Source: research report, Eq. 17

1 · AMPLITUDE

+10.10 rad

per unit ln u_0

Why: more vibration raises the push everywhere, so grip takes longer to catch up.

2 · TENSION

−5.56 rad

per unit ln T_1

Why: a tighter held end raises grip more than push, so grip catches up sooner.

3 · ONSET

0.19 mm

smallest u_0 that gives creep

Why: below it, push is weaker than grip even at the held end.

4 · REFERENCE

θ* = π

at u_0 = 0.256 mm, T_1 = 1 N

Why: the measured constants put the crossing exactly there.

5 · FREQUENCY

no effect

from 30 to 60 Hz

Why: push depends on how far the drum moves, not how fast.

KEY POINT Each prediction follows from the law, and nothing is fitted to the measured θ*, so each one could have failed.

Methodology: experimental apparatus and how it works

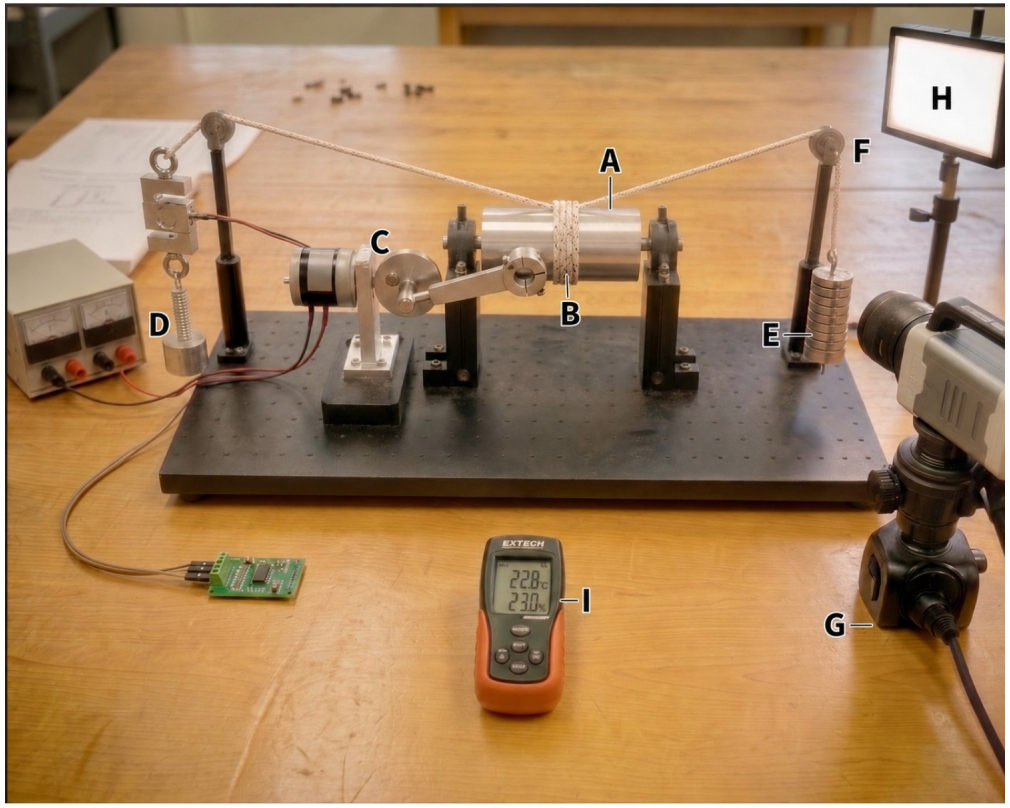


Figure 7 · The apparatus, components A–I
Source: research report, Fig. 3

How the experiment works

A, B A braided polyester rope (B, 3 mm) is wrapped four full turns round an aluminium drum (A, 25 mm diameter).

D, E, F The held end is pulled gently, $T_1 \approx 1$ N, measured by a load cell (D). Weights (E), via pulleys (F), pull the other end with $T_2 = 5\ T_1$.

C A motor, cam and crank (C) twist the drum back and forth, so its surface moves $\pm u_0 \approx 0.26$ mm, 30 to 60 times per second.

G, H, I A high-speed camera (G, 1000 frames s^{-1}), with LED lighting (H) and two mirrors behind the drum, films all four turns. A thermo-hygrometer (I) logs room conditions.

KEY POINT The drum vibrates under fixed end tensions while the camera records how every part of the rope moves.

Methodology: independent calibration of the four constants

μs 0.30 ± 0.02	**Static friction coefficient** The most friction the contact can give before sliding starts. Sets the grip.	**How measured** Half-wrap test: rope over half the still drum; load raised until it slides.
μred 0.18 ± 0.02	**Slow-sliding friction coefficient** Sets how fast tension, and so grip, builds along the rope.	**How measured** Same test, during steady slow sliding.
κref 1.60 ± 0.12 kN m^{-1} rad^{-1}	**Contact stiffness at 1 N** Sideways force per millimetre the drum moves, before slipping. Sets the push.	**How measured** 30° contact; small extra force on one end; recoverable shift measured.
α 0.45 ± 0.05	**Stiffness exponent** How fast stiffness rises with tension: κt grows as T to the power α.	**How measured** Stiffness test repeated at five tensions; power-law fit.

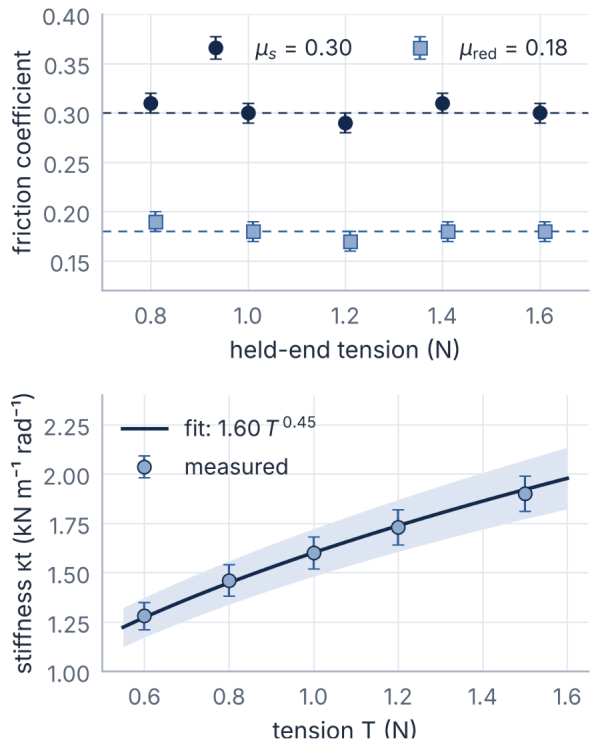


Figure 8 · Friction (top) and stiffness (bottom) tests
Source: redrawn from report Fig. 6(a, b)

KEY POINT All four constants in Equation 1 were measured in separate bench tests before the main experiment.

Experimental design: 14 conditions × 6 repeats = 84 runs

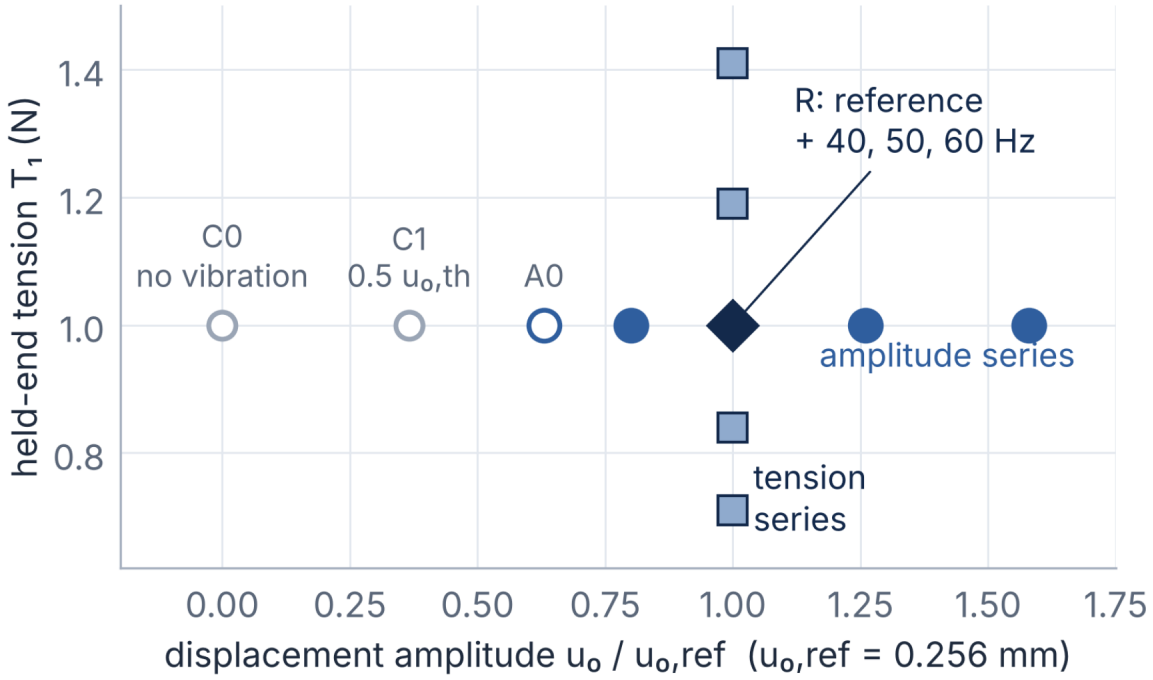


Figure 9 · The 14 conditions: each marker is one setting of vibration amplitude u_o and held-end tension T_1

Source: drawn from report Table 4

What we varied, one at a time, and why

- **Amplitude series** (A0–A3, R): u_o from 0.16 to 0.40 mm at T_1 = 1 N. Tests predictions 1 and 3.
- **Tension series** (T0–T3, R): T_1 from 0.71 to 1.41 N at u_o = 0.256 mm. Tests prediction 2.
- **Frequency**: 30, 40, 50 and 60 Hz at the reference R. Tests prediction 5.
- **Controls** C0 and C1: no vibration, and half the onset amplitude. Test the method itself.

Held fixed

- $T_2 = 5\ T_1$ in every run
- A fresh stretch of rope and drum for every run
- Random run order, so slow drifts cannot pose as trends

KEY POINT Each series changes one variable and tests one prediction; R, the reference setting, is shared by all three series.

Experimental rigour: control conditions

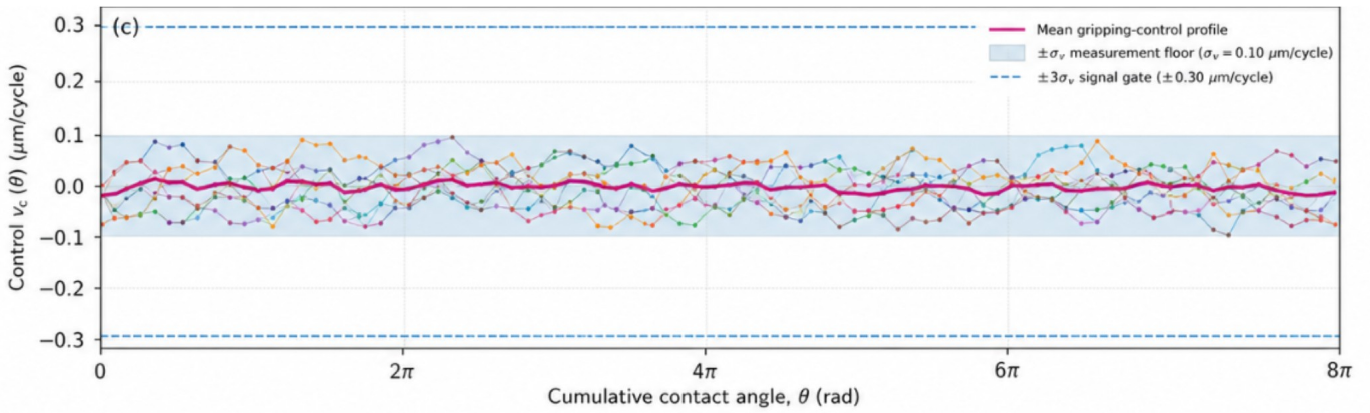


Figure 10 · Creep rate along the rope in the 12 control runs: all flat, at the noise level
Source: reproduced from research report, Fig. 6(c)

✓ Ideal result

False transitions detected

0 of 12

Zero is the target: no transition was reported where creep is physically impossible. All profiles stayed within ±0.10 µm per cycle.

C0 · No vibration (u_o = 0)

Six runs with the drive off. No creep is physically possible, so any transition found would be an error of the method.

C1 · Sub-threshold (u_o = 0.5 u_o,th)

Six runs at half the predicted onset (0.094 mm). The drive runs, but no creep is predicted.

KEY POINT Zero false transitions is the best possible outcome: the method does not report a boundary where none exists.

Step 1 · Measuring how fast each part of the rope creeps

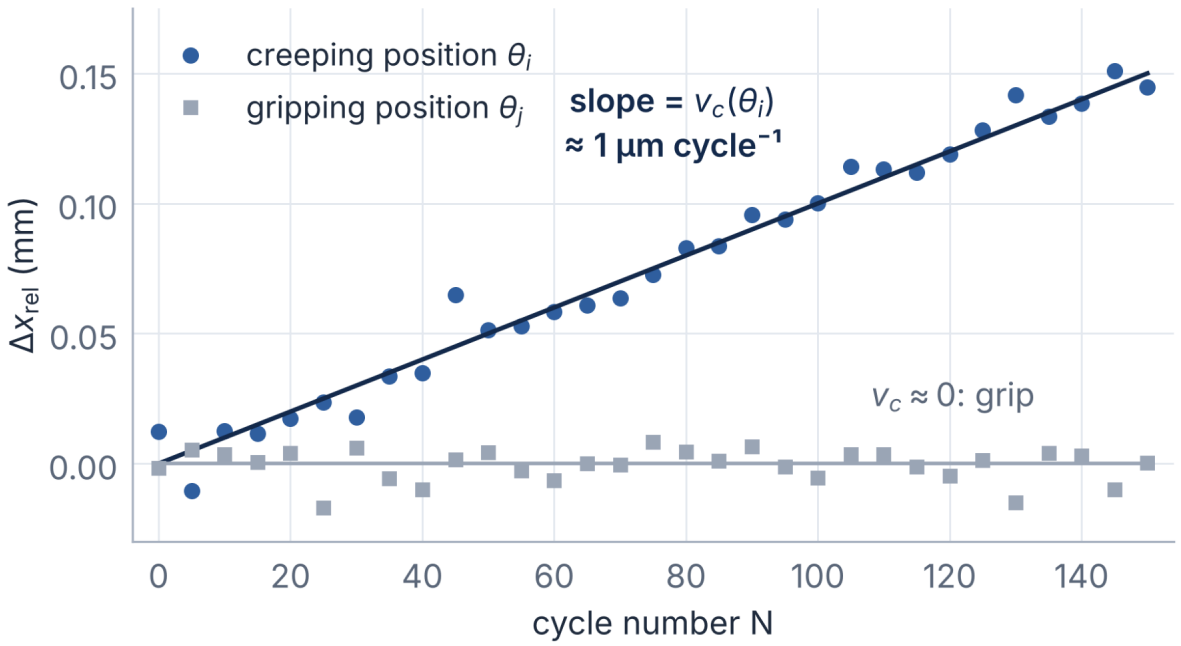


Figure 11 · Slip of the rope relative to the drum, against cycle number, at two positions
Source: illustrative schematic after report Fig. 4(d)

1 Film rope and drum

At 1000 frames per second. Marks on the drum show how the drum moves; the braided texture of the rope shows how each part of the rope moves.

2 Compare at the same moment of each cycle

Within every cycle the rope stretches and springs back with the drum. To remove this, we compare rope and drum only at the same point of the cycle, every 5th cycle, like photographing a swing always at the top.

3 Fit a straight line

Where the rope creeps, its slip relative to the drum grows steadily (blue). The slope is the creep rate, vc, in µm per cycle. Where it holds, the slip stays at zero (grey).

KEY POINT The result is a creep rate vc, in micrometres per cycle, at every position along all four turns of rope.

Step 2 · The creep profile along the rope, and how θ* is read from it

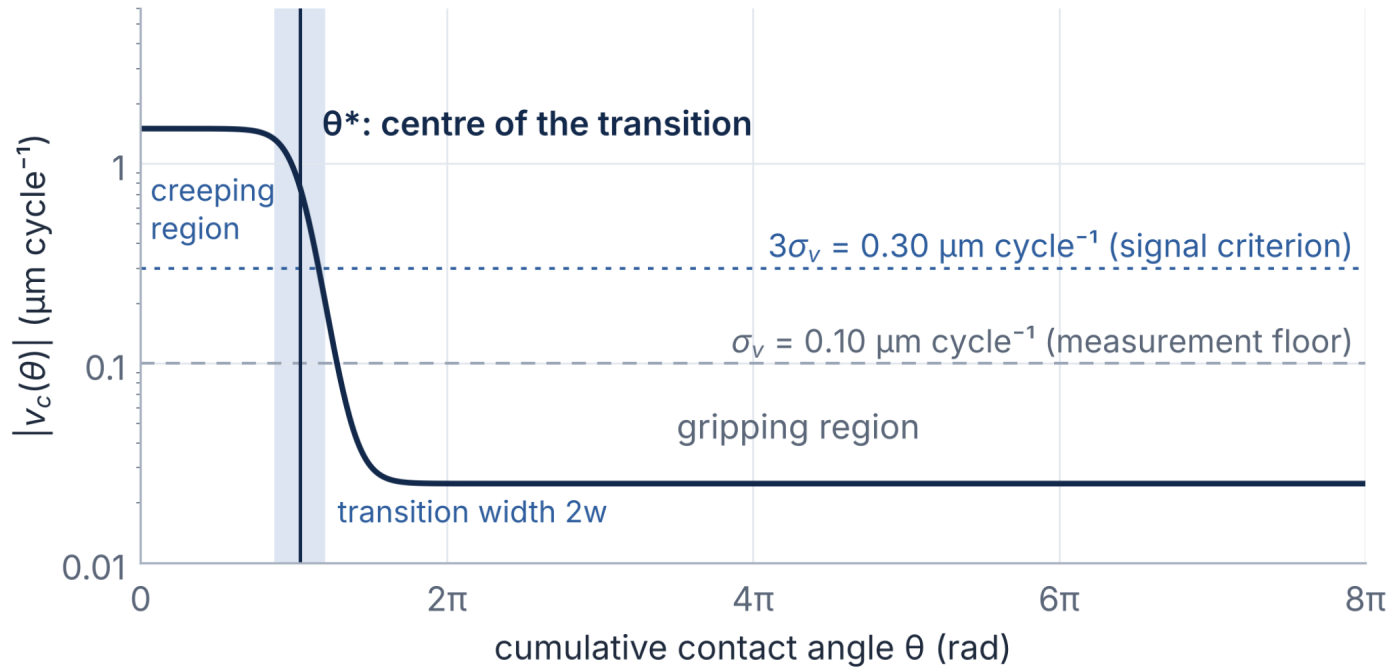


Figure 12 · Creep rate along the whole rope, held end on the left (logarithmic scale)
Source: schematic after report Fig. 4(e), shaped like the reference condition

What the graph shows

The creep rate at every position from Step 1, from the held end (left) to the loaded end (right).

Creep, then grip

High creep near the held end; a short drop, about 13 mm wide; then grip, below the smallest creep rate we can measure (σv = 0.10 µm per cycle).

How θ* is defined

θ* is the middle of the drop. Unlike its edge, the middle does not depend on where the “zero” line is drawn, so it is the most reproducible measure.

KEY POINT θ* is measured as the middle of the drop from creep to grip.

Step 3 · Model selection: is the transition real?

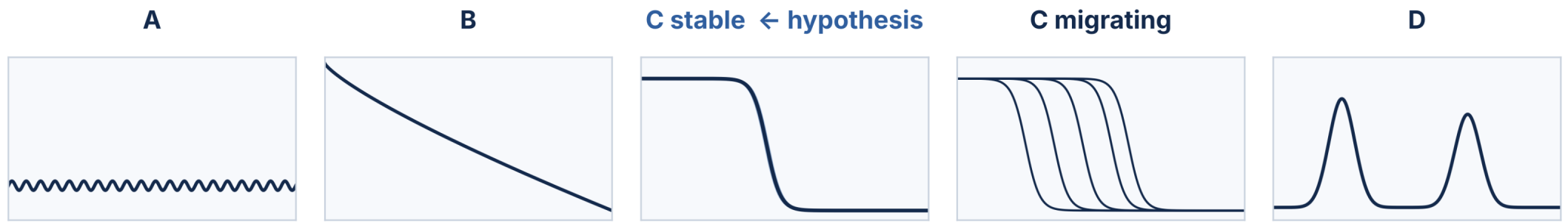


Figure 13 · Five possible shapes for each creep profile; only “C stable” is a single fixed transition
Source: schematic after report Fig. 5 and Table 1

Why this step is needed

Noise can look like a step. So every profile is fitted with all five shapes, and compared using BIC: a score that rewards a close fit but penalises every extra adjustable parameter, so a more flexible shape must fit much better to win.

A transition is accepted only if (rules fixed in advance)

1. “C stable” beats every other shape by ΔBIC ≥ 10, the standard level for very strong evidence
2. The drop is at least 3 times the noise level (0.30 µm per cycle)
3. The transition is in the same place in all six repeats

KEY POINT A transition is reported only when one fixed step clearly beats every alternative explanation.

Result 1 · A single, stable creep–grip transition, as predicted

✓ Matches prediction

Reference setting: measured θ*

3.27 ± 0.17 rad

Predicted: π = 3.14 rad. Difference 0.13 rad (≈ 1.6 mm), inside the uncertainty.

All 14 conditions

- 11 conditions: one stable transition, accepted by all three rules
- No transition in exactly the 3 conditions predicted to have none: C0 (no vibration), C1 and A0 (below the onset)
- 0 of 12 control runs: no false transition

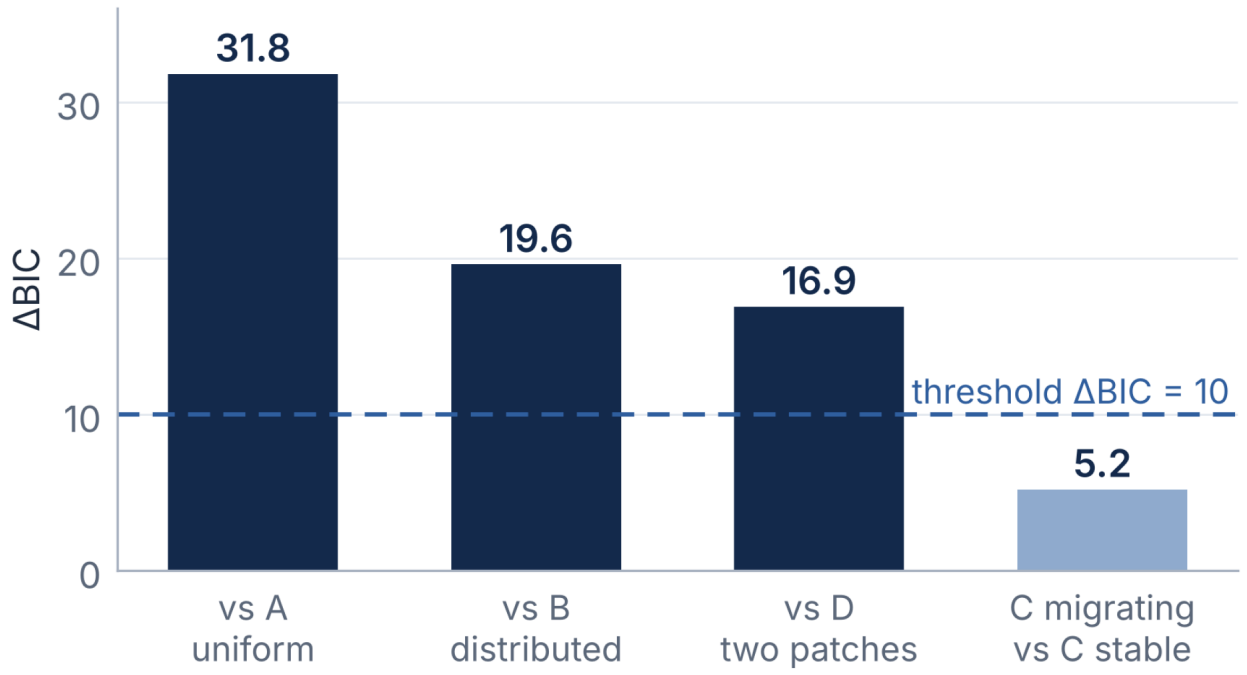


Figure 14 · How strongly "C stable" beats each alternative (above 10 = very strong). Last bar below 10: the transition does not move between repeats, the ideal outcome.
Source: report Section 4.2, Eq. 77

KEY POINT Creep stops at one fixed position, and appears only where the law says it should.

Result 2 · Stronger vibration moves θ* further along the rope

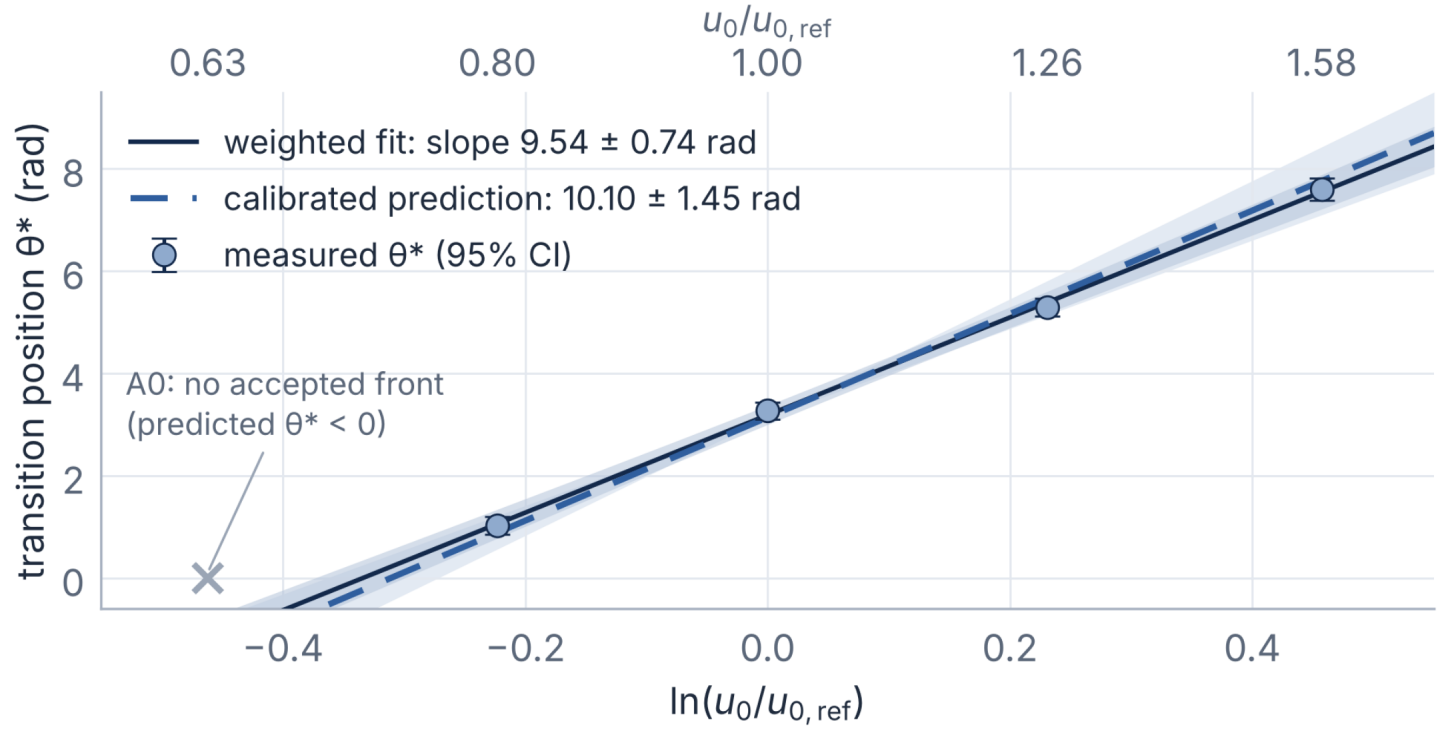


Figure 15 · θ* against vibration amplitude (log scale); solid: fit to the data; dashed: prediction
Source: redrawn from report Fig. 7

✓ Agrees with prediction

Measured slope

9.54 ± 0.74 rad

per unit ln u_0 (fit to the 4 points)

Predicted slope, fixed beforehand

10.10 ± 1.45 rad

= 1/[(1 − α) μred], bench constants only

How to read the graph

Each point is one amplitude. Each 26% rise in u_0 moved θ* about 2.2 rad (28 mm) further along.

KEY POINT Only the vibration amplitude was changed, and θ* moved by the amount the law predicted in advance.

Result 3 · A tighter held end moves θ* back towards it

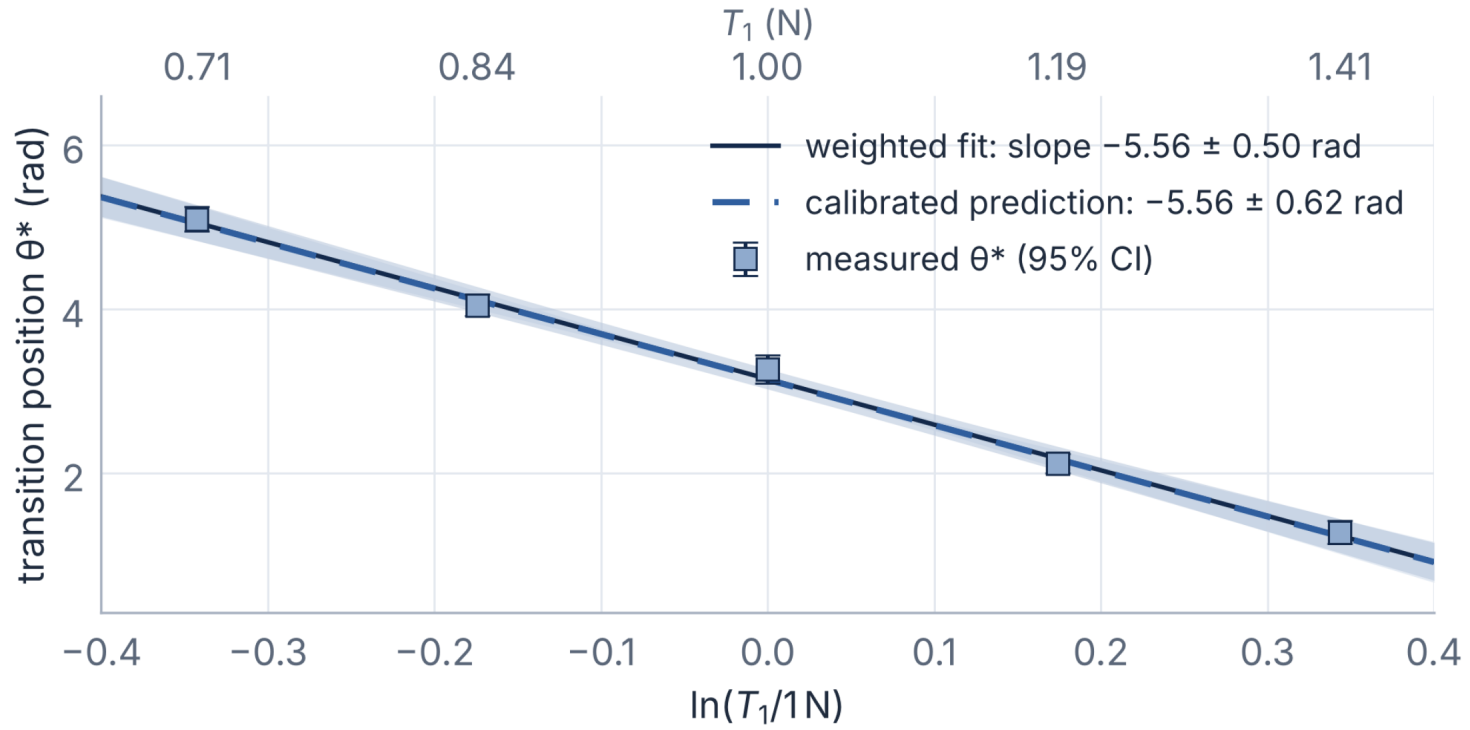


Figure 16 · θ* against held-end tension (log scale); solid: fit to the data; dashed: prediction
Source: redrawn from report Fig. 8

✓ Matches prediction

Measured slope

−5.56 ± 0.50 rad

per unit ln T_1 (fit to the 5 points)

Predicted slope, fixed beforehand

−5.56 ± 0.62 rad

= −1/μred, friction test only

How to read the graph

Each point is one held-end tension. Each 19% increase in T_1 moved θ* about 0.97 rad (12 mm) back towards the held end.

KEY POINT Only the held-end tension was changed, and θ* moved exactly as predicted from the friction test alone.

Result 4 · Predicted against measured θ* for every condition

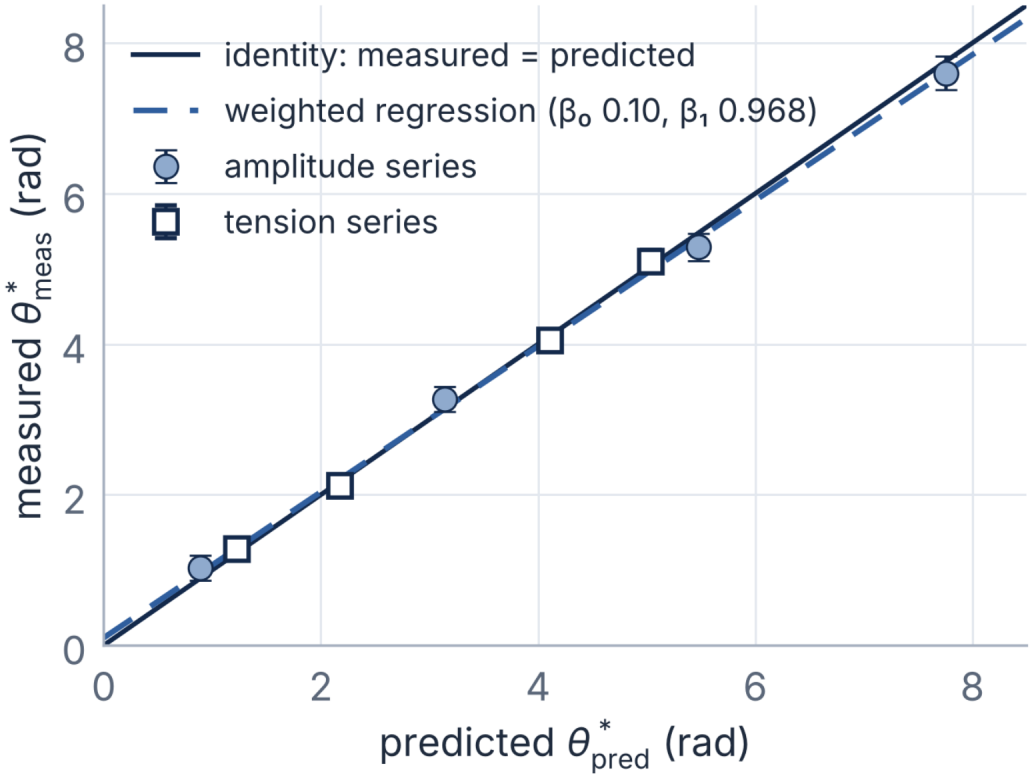


Figure 17 · Measured θ* against θ* predicted before measurement
Source: redrawn from report Fig. 9

✓ Within measurement uncertainty

Typical prediction error (RMSE)

0.117 rad

≈ 1.5 mm of rope, on a contact ≈ 314 mm long

How to read the graph

Each point is one condition; a perfect prediction lies on the diagonal. Best-fit slope 0.968 ± 0.035 (ideal 1); intercept 0.10 ± 0.13 rad (ideal 0).

Onset (prediction 3)

Predicted 0.1875 mm lies between the largest amplitude with no transition (0.161 mm) and the smallest with one (0.205 mm).

KEY POINT The law predicts not only trends but the actual position of the transition, to within about 1.5 mm of rope.

Result 5 · Drive frequency has no effect on θ*, as predicted

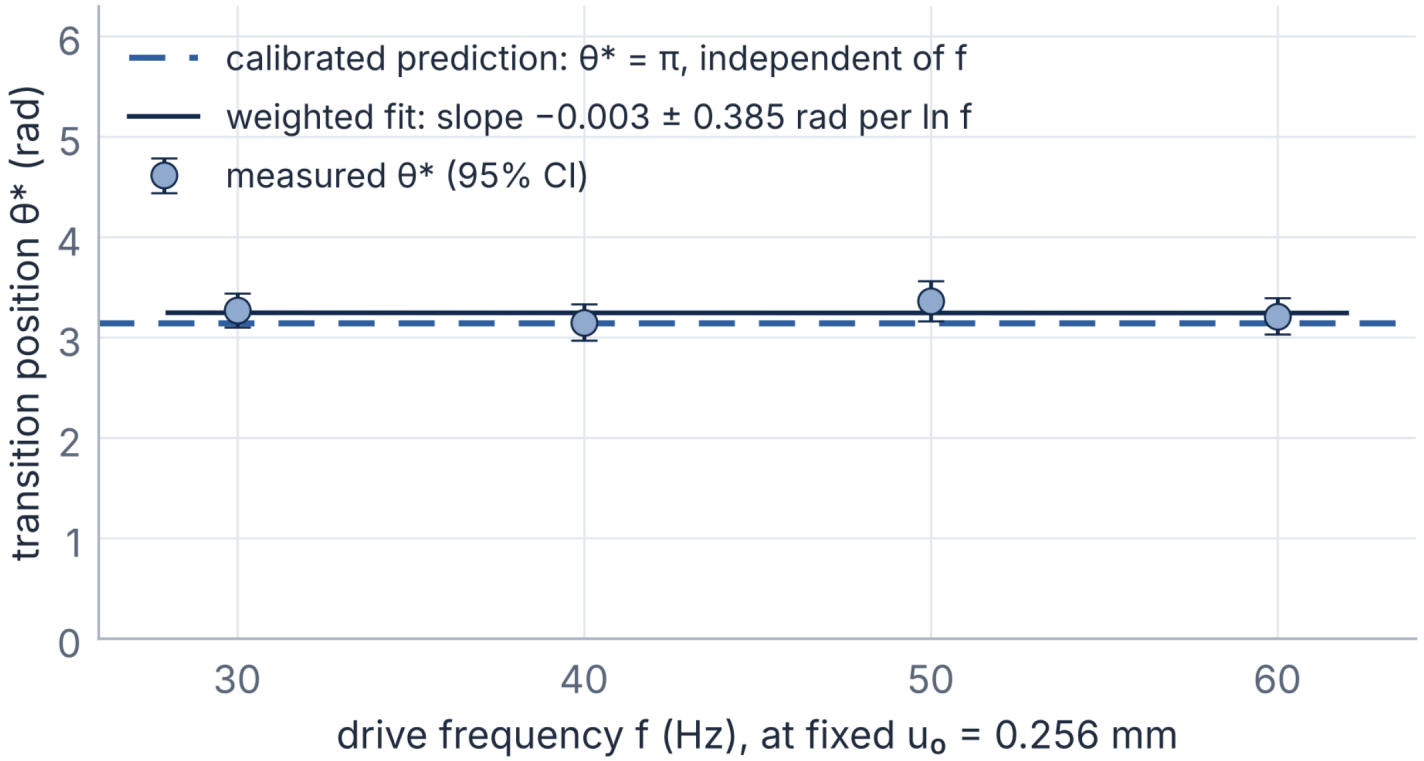


Figure 18 · θ* at four drive frequencies, all at the same amplitude; dashed: prediction θ* = π
Source: redrawn from report Fig. 10

✓ Matches prediction

Effect of frequency on θ*

−0.003 ± 0.385

rad per unit ln f. The law predicts exactly zero, so a value indistinguishable from zero is the confirming result.

How to read the graph

Each point is one frequency, with the drum moving the same distance each time. Doubling the frequency left θ* unchanged (3.15 to 3.36 rad).

KEY POINT θ* depends on how far the drum moves, not how fast: a flat line here is the confirming result.

Validation: recovering the bench constants from the data

The tension slope gives μred back

$$\mu_{red} = -\frac{1}{s_T}$$

μred = 0.180 ± 0.016

bench friction test: 0.18 ± 0.02 ✓

The ratio of the two slopes gives α back

$$\alpha = 1 + \frac{s_T}{s_u}$$

α = 0.417 ± 0.069

bench stiffness test: 0.45 ± 0.05 ✓

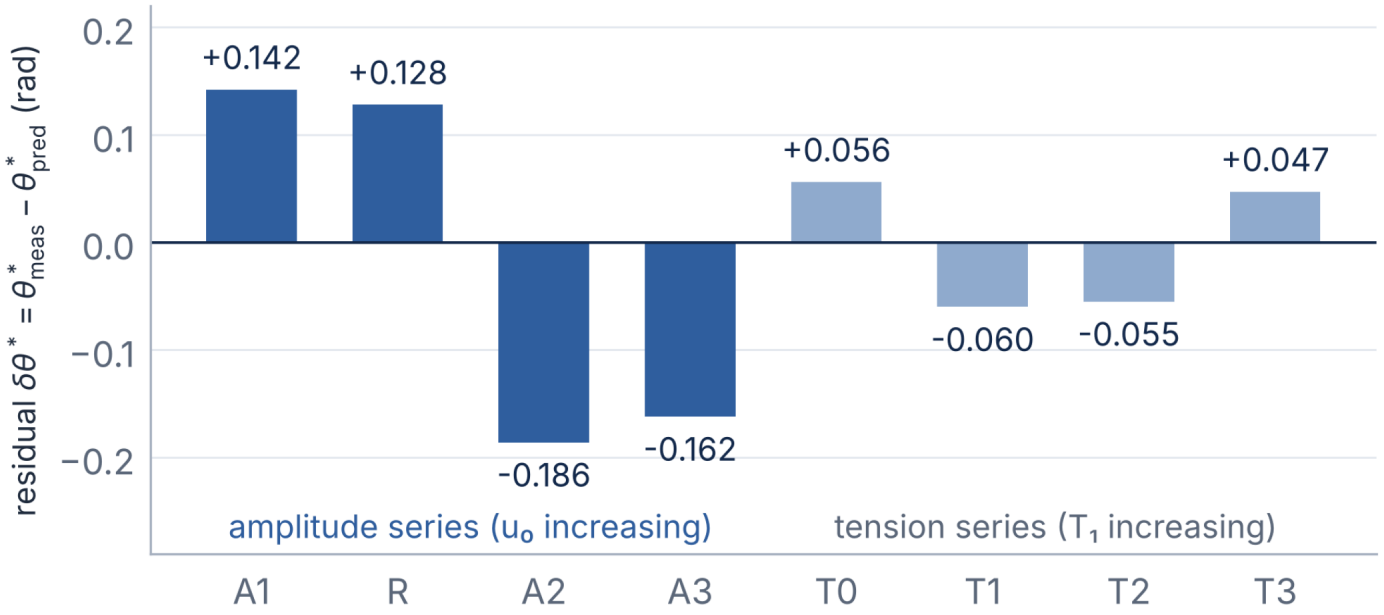


Figure 19 · Measured minus predicted θ* for each condition: all within ±0.19 rad (±2.4 mm)
Source: research report, Eq. 110

KEY POINT Working backwards from the creep data returns the same constants that the separate bench tests measured.

Limitations and future work

Limitation	Implication and future work
The reduced resistance profile Teff is an effective closure, not the true tension distribution	The model targets only the transition centre; its validity rests on agreement with experiment
Additive per-radian stiffness assumed along the wrap	Verified only on short contacts of this rope; to be tested on longer arcs
T_1 and T_2 co-varied ($T_2/T_1 = 5$)	Independent T_1 dependence not isolated. Next: T_2 sweep at fixed T_1
Single rope–capstan pair; 30–60 Hz; small amplitudes	Next: other materials, wider frequency and amplitude ranges
Creep measured, wear not quantified	Next: correlate the creep region with measured fibre damage

KEY POINT **Within the tested parameter range, every conclusion held under all robustness diagnostics.**

Potential impact and applications

SCIENTIFIC IMPACT

From load limit to position

Capstan theory gains a spatial prediction: the location of vibration-driven creep.

Transferable framework

For any wrapped contact, measure μ and κt(T) and θ* follows before the system is built.

ENGINEERING APPLICATIONS

Winches and mooring lines

Targeted inspection of the creep region.

Cable-driven robots

Locating cable drift and wear.

Capstan brakes, textile guides

Predicting band and yarn wear location.

SOCIETAL IMPACT: OFFSHORE MOORING SAFETY

Cost of one mooring failure

$1.8 billion

to reinstate the Gryphon Alpha FPSO after 4 of its 10 mooring lines parted (North Sea, 2011)

Failures are rarely foreseen

> 50% surprises

of mooring-failure root causes, 2001–2011; chain failures cluster at the fairlead, where the line wraps a contact

Our contribution

θ* makes the wear location predictable: which stretch of a wrapped line creeps and wears under vibration, from bench tests, before installation.

KEY POINT Lines fail where they wrap and rub, often without warning; this law predicts where that is.

Conclusion

ANSWER TO THE RESEARCH QUESTION

Vibration-driven creep does not extend over the whole wrap. It initiates at the held end and terminates at a localized transition θ*, whose position is predicted by a closed-form law built entirely from independently calibrated friction and stiffness parameters.

Transition
11 of 14 conditions; 0 of 12 controls

Amplitude sensitivity
9.54 measured, 10.10 predicted

Tension sensitivity
−5.56 measured, −5.56 predicted

Absolute position
RMSE 0.117 rad ≈ 1.5 mm

Frequency
no dependence, 30–60 Hz

Parameter recovery
μred = 0.180
calibration: 0.18

Parameter recovery
α = 0.417
calibration: 0.45

Robustness
6 of 6
stability diagnostics within limits

KEY POINT Capstan friction, combined with preslip contact stiffness, predicts not only a load limit but a position.

Thank you

Questions are welcome · Presented by Jevesh Jain

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BACK-UP MATERIAL FOR QUESTIONS

Supplementary slides

S1 · Regimes of the law

S2 · Predictions with their equations

S3 · Phase-matched sampling and extraction workflow

S4 · BIC model selection and acceptance rules

S5 · Run protocol and quality control

S6 · Robustness of the results

S7 · Summary of all 84 runs

Main theoretical result: the predictive law for θ*

loading-to-resistance ratio at the held end, $q_{vib}(0)/q_{res}(0)$

$$\theta^* = \frac{1}{(1-\alpha)\,\mu_{red}} \ln\left(\frac{K_{ref}\,u_0}{\mu_s\,T_{ref}^{\alpha}\,T_1^{1-\alpha}}\right)$$

inverse of the net growth rate of resistance over loading

Equation 1 · Transition-position law
Source: research report, Eq. 17

Sub-threshold: $u_0 < u_{0,th}$

Loading is below resistance everywhere: the logarithm is negative and no creep region forms.

Partial mobilisation (tested)

Creep for $0 < \theta < \theta^*$, grip beyond: $0 < \theta^* < \Theta$.

Full mobilisation: $u_0 > u_{0,full}$

Loading exceeds resistance over the entire wrap: the whole contact creeps.

KEY POINT Every parameter is calibrated independently; no quantity is adjusted to fit the measured θ*.

Testable predictions

1 · AMPLITUDE SENSITIVITY

$$\frac{\partial \theta^*}{\partial \ln u_0} = \frac{1}{(1-\alpha)\,\mu_{red}} = 10.10$$

Test: amplitude series at fixed T_1 (±1.45)

2 · TENSION SENSITIVITY

$$\frac{\partial \theta^*}{\partial \ln T_1} = -\frac{1}{\mu_{red}} = -5.56$$

Test: tension series at fixed u_0 (±0.62)

3 · ONSET THRESHOLD

$$u_{0,\mathrm{th}} = \frac{\mu_s\, T_{ref}^{\alpha}\, T_1^{1-\alpha}}{\kappa_{ref}} = 0.1875 \text{ mm}$$

Test: no accepted front below $u_{0,th}$

4 · ABSOLUTE POSITION

θ*ref = π

at the reference condition; a numerical prediction for every condition

Test: θ*meas against θ*pred

5 · FREQUENCY INDEPENDENCE

∂θ*/∂ln f = 0

frequency does not enter Eq. 17

Test: fixed u_0 at 30, 40, 50, 60 Hz

KEY POINT Each prediction was fixed before the confirmatory runs, so each could have falsified the law.

Phase-matched sampling and the extraction workflow

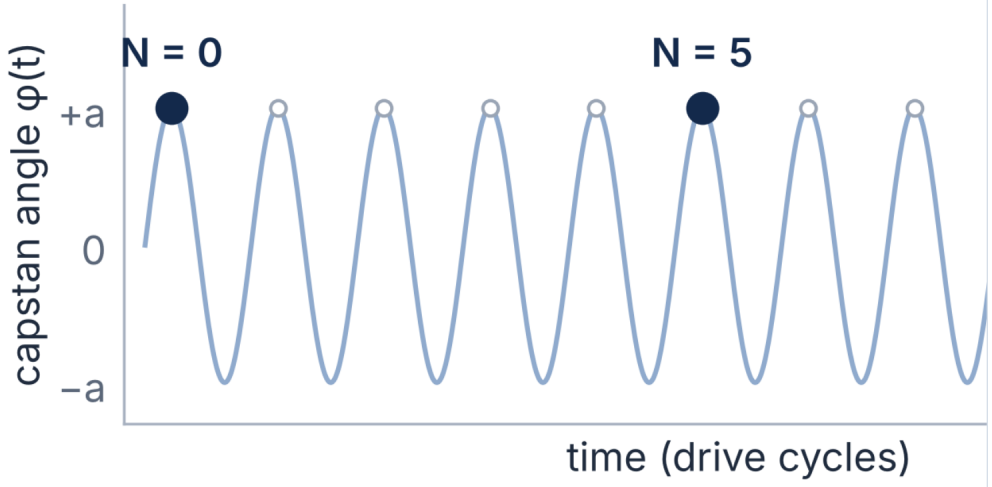


Figure 20 · Phase-matched sampling at equivalent mechanical sta

Source: schematic after report Fig. 4(b, c)

Extraction workflow (report Eq. 115)

1. Record at 1000 frames s^{-1}; direct and mirror views cover all four wraps.
2. Track a drum-fixed feature to obtain the drum angle φ(t); its positive extrema define the phase reference.
3. Track the rope's braid texture at many positions; no interpolation.
4. Register all views onto one coordinate, θ = 0 to 8π.
5. Sample rope and drum at matched phase every 5th cycle, N = 0–150.

KEY POINT Phase matching removes the reversible elastic motion, isolating the slip that accumulates from cycle to cycle.

Step 5 · Model selection and acceptance criteria

BAYESIAN INFORMATION CRITERION (EQ. 58)

$$\mathrm{BIC} = n \ln\left(\frac{\mathrm{RSS}}{n}\right) + k \ln n$$

Goodness of fit
n data points; RSS: residual sum of squares

Complexity penalty
k: number of free parameters

Decision rule
lower BIC preferred; ΔBIC ≥ 10 is very strong evidence

Front acceptance criteria (pre-set)

1. C stable preferred over A, B and D by ΔBIC ≥ 10
2. Front amplitude exceeds 3σv = 0.30 μm cycle^{-1}
3. θ* at least 0.35 rad from either boundary
4. C migrating does not improve BIC by ≥ 10

KEY POINT Acceptance criteria were fixed before analysis, so the classification could not be tuned to the result.

Experimental rigour: run protocol and quality control

TIMED SEQUENCE FOR EVERY RUN (REPORT EQ. 113)

60 s	→	5 s	→	1 s	→	2 s	→	5 s	→	5 s
settling		T_1 pre-measurement		static baseline		drive spin-up		high-speed recording		T_1 post-measurement

Run acceptance (report Eq. 114)

- Tension stability: $|T_{1,}\text{post} - T_{1,}\text{pre}| \,/\, T_{1,}\text{pre} \le 2\%$
- Amplitude stability: $|\Delta u_0| \,/\, u_0 \le 3\%$
- Runs failing either criterion are excluded before analysis

Analysis consistency

- Common window N = 0–150 at every frequency
- Phase-matched samples every 5th cycle (31 per position)
- All quoted uncertainties are 95% intervals

KEY POINT A fixed protocol and pre-set acceptance criteria ensure every run is directly comparable.

Statistical rigour: robustness of the results

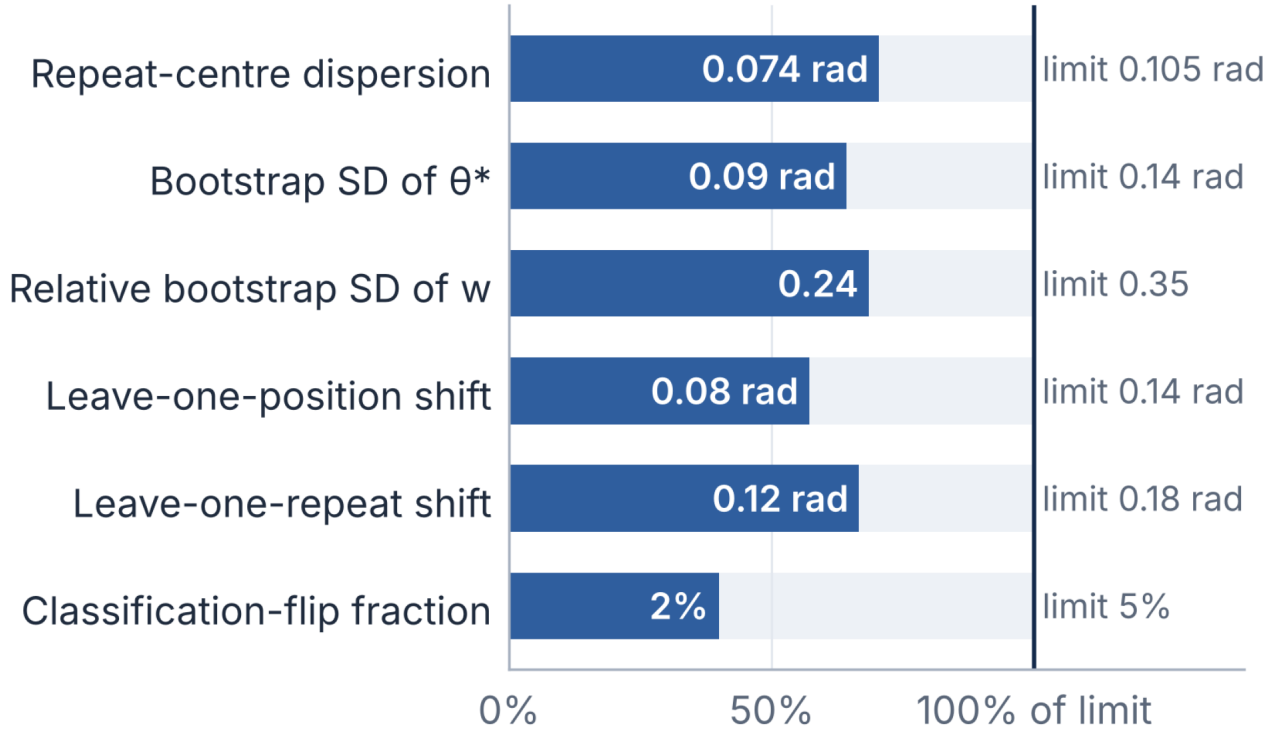


Figure 21 · Largest observed instability across the 11 accepted fronts, relative to its pre-set limit
Source: data from report Table 3

Bootstrap resampling

Repeat-level and block resampling; full re-analysis of each resampled dataset.

Deletion stability

Leave-one-repeat-out and leave-one-position-out refits.

Repeat consistency

Independent fits of each repeat; dispersion of the six centres.

KEY POINT Every diagnostic remained within its pre-set limit: no single repeat or position determines θ*.

Summary of results: all 84 runs

Condition	u_0 (mm)	T_1 (N)	f (Hz)	θ*pred (rad)	θ*meas (rad)	Classification
A0	0.161	1.00	30	none (−1.53)	—	No front, as predicted
A1	0.205	1.00	30	0.89	1.03 ± 0.17	Stable front
R	0.256	1.00	30	3.14	3.27 ± 0.17	Stable front
A2	0.322	1.00	30	5.48	5.29 ± 0.18	Stable front
A3	0.404	1.00	30	7.76	7.60 ± 0.22	Stable front
T0	0.256	0.71	30	5.04	5.10 ± 0.15	Stable front
T1	0.256	0.84	30	4.11	4.05 ± 0.13	Stable front
T2	0.256	1.19	30	2.18	2.12 ± 0.12	Stable front
T3	0.256	1.41	30	1.23	1.28 ± 0.14	Stable front
F40	0.256	1.00	40	3.14	3.15 ± 0.18	Stable front
F50	0.256	1.00	50	3.14	3.36 ± 0.20	Stable front
F60	0.256	1.00	60	3.14	3.21 ± 0.18	Stable front
C0	0	1.00	—	none	—	No front, as predicted
C1	0.094	1.00	30	none (−7.00)	—	No front, as predicted

Table 1 · All 14 conditions, 6 repeats each, $T_2 = 5\ T_1$; a negative formal prediction denotes a predicted no-front condition · Source: research report, Table 4

KEY POINT Classification matched prediction in all 14 conditions: 11 fronts where predicted, none where not.